STPM 2026 PHYSICS 960/2

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}.

Section A [20 marks]

1. Two particles with charges, Q and $-2Q$, are placed at a fixed distance. Which statement is true about the electrostatic force that acts between the charges?

A The attractive force with the same magnitude.
B The repulsive force with the same magnitude.
C The attractive force that acts on Q by $-2Q$ is twice the attractive force that acts on $-2Q$ by Q.
D The repulsive force that acts on Q by $-2Q$ is twice the repulsive force that acts on $-2Q$ by Q.

Answer: A

Unlike charges attract. By Newton's third law the force on Q due to $-2Q$ and the force on $-2Q$ due to Q are equal in magnitude and opposite in direction.

Explanation: Coulomb's law gives one force, $F = \frac{Q(2Q)}{4\pi\varepsilon_0 r^2}$, acting on each charge. The larger charge does not feel a larger force.

2. The electric field strength and the electric potential on the surface of a solid metal sphere which has a fixed charge are E_o and V_o respectively. Which is the electric field strength and electric potential at a point inside the sphere?

	Electric field strength	*Electric potential*
A	0	0
B	0	V_o
C	E_o	V_o
D	E_o	0

Answer: B

Inside a charged conductor the electric field is zero, so no work is done moving a charge inside it: the potential everywhere inside equals the surface potential V_o.

Explanation: $E = -\mathrm{d}V/\mathrm{d}r$: zero field means constant (not zero) potential.

3. A parallel-plate capacitor with a plate area of A and a plate separation of d is charged in air with charge Q. If the potential difference across the capacitor is V and the electric field strength between the plates of the capacitor is E, the capacitance of the capacitor is

A QEd B $\frac{AE}{d}$ C $\frac{Ed}{A}$ D $\frac{Q}{Ed}$

Answer: D

$$C = \frac{Q}{V} \text{ and } V = Ed \Longrightarrow C = \frac{Q}{Ed}$$

Explanation: In a uniform field between parallel plates, the potential difference is the field strength times the separation.

4. A dielectric material is inserted between the plates of a parallel-plate capacitor. Which statement is true about the effect of the polarisation in the capacitor?

A The charge stored increases.
B The effective capacitance decreases.
C The effective electric field decreases.
D The potential difference between the plates increases.

Answer: C

The polarised molecules of the dielectric produce an induced field opposite to the applied field, so the effective (resultant) electric field between the plates decreases.

Explanation: For a fixed charge, a smaller field means a smaller p.d. ($V = Ed$) and so a larger capacitance ($C = Q/V$).

5. Which of the following will happen microscopically to a current-carrying conductor when the electric field in the conductor increases?

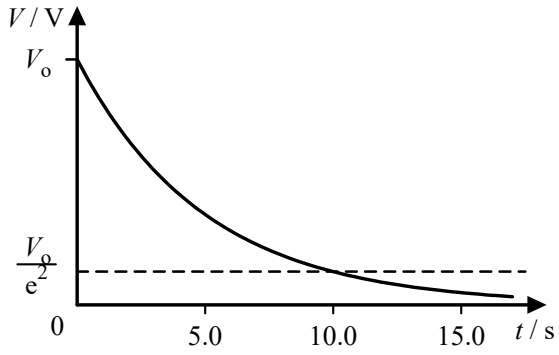
A The current density decreases.
B The drift velocity of the electrons increases.
C The mean free time of the electrons increases.
D The electrons move faster in the direction of the current.

Answer: B

$v_d = \frac{eE}{m}\tau$: a larger field gives a larger drift velocity

Explanation: The mean free time depends on the lattice (temperature), not on the field. Electrons drift opposite to the conventional current. The current density $J = nev_d$ increases.

6. The variation of potential difference, V, with time, t, for a discharging capacitor through a resistor of 2.0 MΩ is shown in the graph below.



What is the capacitance of the capacitor?

A 1.3 μF B 2.5 μF C 3.8 μF D 5.0 μF

Answer: B

$$V = V_o e^{-t/RC};\ V = \frac{V_o}{e^2} \text{ at } t = 10.0 \text{ s} \Longrightarrow t = 2RC$$

$$RC = 5.0 \text{ s} \Longrightarrow C = \frac{5.0}{2.0 \times 10^6} = 2.5 \text{ μF}$$

Explanation: After one time constant the p.d. falls to V_o/e; after two time constants to V_o/e^2.

7. A conductor of cross-sectional area of 5.0×10^{-6} m^2 and length of 2.0 m contains 4.17×10^{23} free electrons. When the conductor is connected to a battery, the drift velocity of the conduction electrons is 1.2×10^{-4} m s^{-1}. What is the current density in the conductor?

A 2.0×10^{-5} A m^{-2} B 4.0×10^{5} A m^{-2} C 8.0×10^{5} A m^{-2} D 1.6×10^{6} A m^{-2}

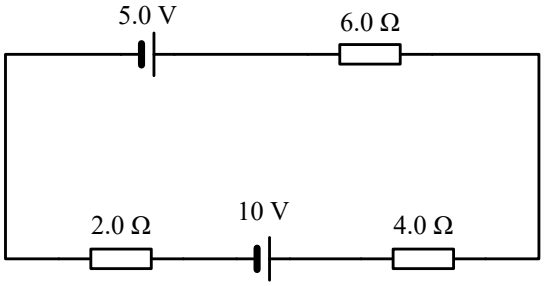
Answer: C

$$n = \frac{4.17 \times 10^{23}}{(5.0 \times 10^{-6})(2.0)} = 4.17 \times 10^{28} \text{ m}^{-3}$$

$$J = nev_d = (4.17 \times 10^{28})(1.60 \times 10^{-19})(1.2 \times 10^{-4}) = 8.0 \times 10^{5} \text{ A m}^{-2}$$

Explanation: n is the number of free electrons per unit volume, so divide the total number by the volume Al.

8. A circuit consists of three resistors of resistance 2.0 Ω, 4.0 Ω and 6.0 Ω respectively, and two batteries of e.m.f.s, 5.0 V and 10 V, as shown in the circuit diagram below.



Which is the magnitude and direction of the current in the circuit?

	Magnitude	*Direction of current*
A	1.25 A	Clockwise
B	0.42 A	Clockwise
C	1.25 A	Anticlockwise
D	0.42 A	Anticlockwise

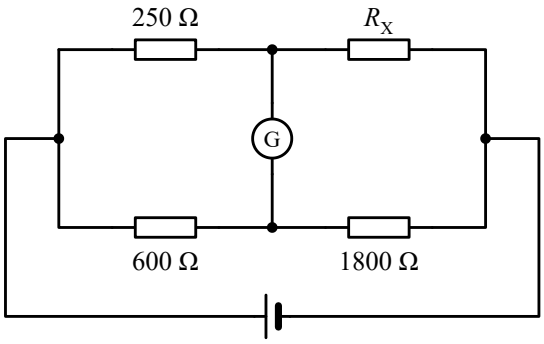
Answer: D

The two batteries oppose each other: net e.m.f. = 10 − 5.0 = 5.0 V, in the direction of the 10 V battery.

$$I = \frac{5.0}{2.0 + 4.0 + 6.0} = 0.42 \text{ A, anticlockwise}$$

Explanation: When cells in a single loop face opposite ways, subtract their e.m.f.s; the current flows in the direction driven by the larger e.m.f.

9. Four resistors of resistance 250 Ω, 600 Ω, 1800 Ω and R_X are arranged as shown in the circuit diagram below.



If the galvanometer reading is zero, which statement is true about the circuit?

A The resistance of R_X is larger than 600 Ω.

B The current flows in the galvanometer is large.

C The current flows in 250 Ω is equal to the current flows in 600 Ω.

D The potential difference across R_X is equal to the potential difference across 1800 Ω.

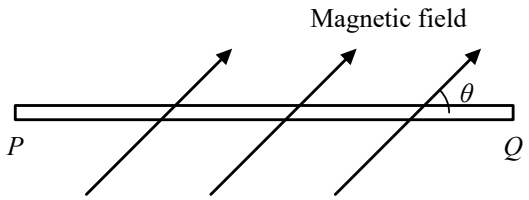
Answer: D

$$\text{Balanced bridge: } \frac{250}{R_X} = \frac{600}{1800} \Longrightarrow R_X = 750\ \Omega$$

No current in G means the two junctions are at the same potential, so the p.d. across R_X equals the p.d. across 1800 Ω (D).

Explanation: D states the balance condition directly. Note that $R_X = 750\ \Omega$ is also larger than 600 Ω, so statement A is true as well; A or D should be accepted.

10. A conducting rod, PQ, of weight 8.0 N and length of 40 cm that carries a current of 20 A is shown in the diagram below.



A uniform magnetic field of 2.0 T is applied into the page at an angle, θ, with the rod. If the magnetic force is just enough to balance the weight of the rod, which is the value of θ and the direction of the current?

	θ	*Direction of current*
A	30°	P to Q
B	30°	Q to P
C	60°	Q to P
D	60°	P to Q

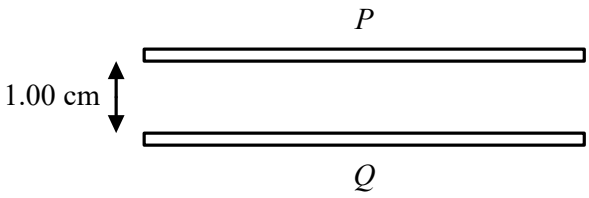
Answer: A

$$BIl \sin\theta = W \Longrightarrow \sin\theta = \frac{8.0}{(2.0)(20)(0.40)} = 0.50 \Longrightarrow \theta = 30°$$

For the force to be upward with the field into the page, the current flows from P to Q (Fleming's left-hand rule).

Explanation: Only the component of the field perpendicular to the rod, $B \sin\theta$, produces a force.

11. Two long straight parallel wires, P and Q, with a separation of 1.00 cm apart is shown in the diagram below.



P carries a current of 30.0 A to the east. If the force per unit length between the wires is 5.00×10^{-2} N m^{-1}, which is the magnitude and direction of the current in Q so that Q does not fall due to gravity?

A 83.3 A to the west
B 83.3 A to the east
C 26.5 A to the west
D 26.5 A to the east

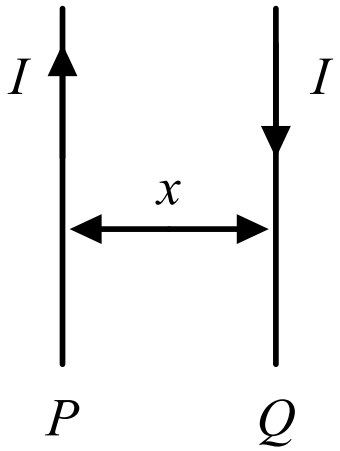
Answer: B

$$\frac{F}{l} = \frac{\mu_0 I_P I_Q}{2\pi d} \Rightarrow I_Q = \frac{(5.00 \times 10^{-2})(2\pi)(0.0100)}{(4\pi \times 10^{-7})(30.0)} = 83.3 \text{ A}$$

Q hangs below P, so the force on Q must be upward (attraction). Parallel currents attract, so the current in Q is to the east.

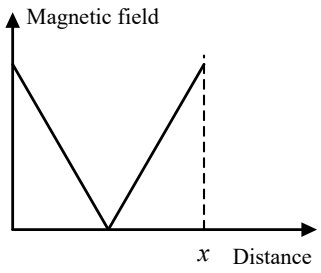
Explanation: Currents in the same direction attract; currents in opposite directions repel. Opposite currents would push Q further down.

12. Two long straight parallel wires, P and Q, carry equal currents in opposite directions as shown in the diagram below.

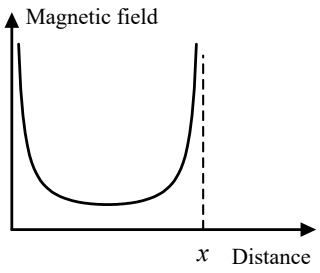


The distance from P to Q is x. Which graph shows the variation of magnitude of the magnetic field with the distance from P to Q?

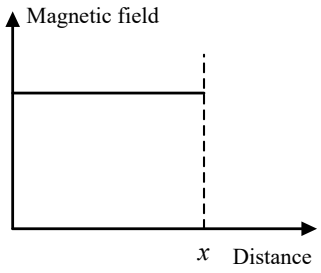
A



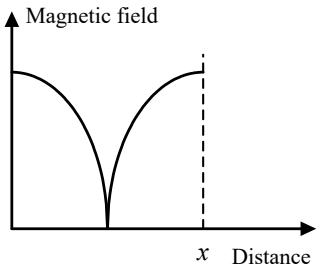
B



C



D



Answer: B

Opposite currents: between the wires both fields point the same way and add. Each field is $\propto 1/r$, so the total is very large near each wire and smallest (but not zero) at the midpoint.

Explanation: If the currents were in the same direction, the fields would cancel at the midpoint, giving zero there.

13. A solenoid of length 3.0 cm is used for an automatic door locking system which is operated by a direct current voltage of 12 V. The magnetic field produced inside the solenoid is 1.5 mT. What is the number of turns of the solenoid if the resistance of the solenoid is 15 Ω?

A 5 turns B 28 turns C 44 turns D 89 turns

Answer: C

$$I = \frac{12}{15} = 0.80 \text{ A}; B = \frac{\mu_0 NI}{l} \Longrightarrow N = \frac{(1.5 \times 10^{-3})(0.030)}{(4\pi \times 10^{-7})(0.80)} \approx 44 \text{ turns}$$

Explanation: Find the current from Ohm's law first, then use the field inside a long solenoid.

14. Which action will increase the magnetic flux through a conducting loop in a uniform magnetic field?

A Rotate the loop about an axis that is parallel to the magnetic field
B Decrease the strength of the magnetic field
C Increase the number of turns in the loop
D Increase the area of the loop

Answer: D

$\Phi = BA \cos\theta$: increasing the area A increases the flux

Explanation: Extra turns increase the flux **linkage** $N\Phi$, not the flux through the loop. Rotating about an axis parallel to **B** can only reduce the flux, and decreasing B reduces it.

15. Which is true about Lenz's law?

A Determines the strength of the magnetic field around a current-carrying wire
B Determines the magnitude of induced e.m.f.
C Dictates the direction of induced current
D Dictates the direction of magnetic force

Answer: C

Lenz's law: the induced current flows in a direction that opposes the change producing it. It gives the direction of the induced current.

Explanation: The magnitude of the induced e.m.f. is given by Faraday's law.

16. A solenoid of length of 5.0 cm and cross-sectional area of 1.5 cm^2 consists of 300 turns. What is the self inductance of the solenoid?

A 1.1×10^{-6} H B 1.7×10^{-5} H C 3.4×10^{-4} H D 5.1×10^{-4} H

Answer: C

$$L = \frac{\mu_0 N^2 A}{l} = \frac{(4\pi \times 10^{-7})(300)^2(1.5 \times 10^{-4})}{0.050} = 3.4 \times 10^{-4} \text{ H}$$

Explanation: Convert the area: 1.5 $cm^2 = 1.5 \times 10^{-4}$ m^2.

17. A primary coil of 200 turns with an inductance of 0.30 H is placed coaxially with a secondary coil of the same cross-sectional area. If the mutual inductance is 0.60 H, what is the number of turns in the secondary coil?

A 100 turns B 120 turns C 200 turns D 400 turns

Answer: D

$$L_1 = \frac{\mu_0 {N_1}^2 A}{l}, M = \frac{\mu_0 N_1 N_2 A}{l} \Rightarrow \frac{M}{L_1} = \frac{N_2}{N_1}$$

$$N_2 = 200 \times \frac{0.60}{0.30} = 400 \text{ turns}$$

Explanation: With the same area and length, the mutual inductance and the self inductance differ only in the number of turns.

18. An alternating voltage supply, $V = 6.0 \sin 50t$, is connected to a pure resistor of resistance 3.0 Ω. What is the average power dissipated by the resistor?

A 6.0 W B 12 W C 18 W D 54 W

Answer: A

$$\langle P \rangle = \frac{V_{\text{rms}}^2}{R} = \frac{V_0^2}{2R} = \frac{6.0^2}{2(3.0)} = 6.0 \text{ W}$$

Explanation: The peak voltage is 6.0 V; use the r.m.s. value for average power.

19. An alternating power supply with r.m.s. voltage of 240 V and frequency of 50.0 Hz is connected in series to a capacitor of capacitance 2200 μF and a resistor of resistance 10.0 Ω. What is the r.m.s. voltage across the capacitor?

A 1.45 V B 23.8 V C 30.3 V D 34.4 V

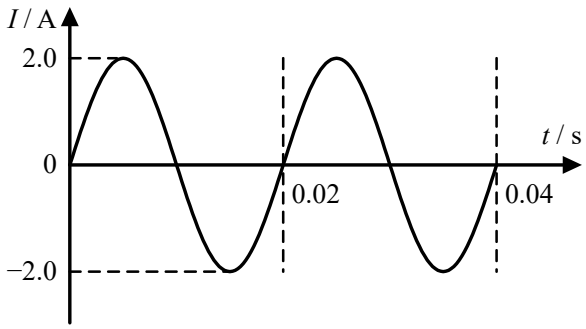
Answer: D

$$X_C = \frac{1}{2\pi(50.0)(2200 \times 10^{-6})} = 1.45 \ \Omega; \ Z = \sqrt{(10.0^2 + 1.45^2)} = 10.1 \ \Omega$$

$$I_{\text{rms}} = \frac{240}{10.1} = 23.8 \text{ A}; \ V_C = IX_C = (23.8)(1.45) = 34.4 \text{ V}$$

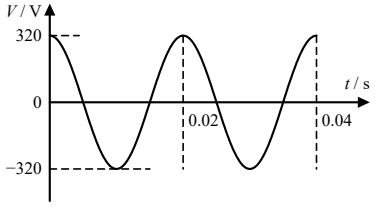
Explanation: Find the circuit current from the impedance, then multiply by the reactance of the capacitor. Option A is the reactance and B is the current.

20. The variation of an alternating current, I, with time, t, for a capacitor of capacitance 20 μF is shown in the graph below.

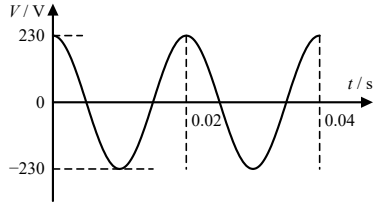


Which graph shows the variation of potential difference, V, across the capacitor with time, t?

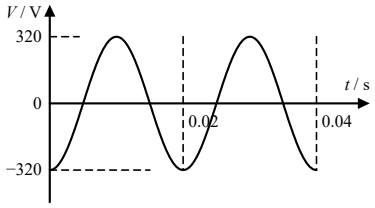
A



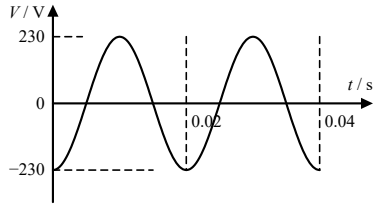
B



C



D



Answer: C

$$T = 0.02\ \text{s} \Longrightarrow \omega = 100\pi\ \text{rad s}^{-1};\ V_0 = I_0 X_C = \frac{2.0}{(100\pi)(20 \times 10^{-6})} = 318\ \text{V} \approx 320\ \text{V}$$

In a capacitor the voltage lags the current by 90°: $I = 2.0 \sin \omega t$ gives $V = -320 \cos \omega t$, starting at −320 V.

Explanation: When the current is zero and about to increase (t = 0), the capacitor is fully charged in the opposite sense, so the p.d. is at its negative peak.

Section B [14 marks]

21. Copper is used in a high voltage transmission line cable which carries a current of 50.0 A. The resistance of the cable is $1.50 \times 10^{-4}\ \Omega\ \text{m}^{-1}$.

(a) Calculate the power dissipated along 300 km of the cable. [2]

Answer

$R = (1.50 \times 10^{-4})(300 \times 10^{3}) = 45.0\ \Omega$ — 1

$P = I^2R = (50.0)^2(45.0) = 1.13 \times 10^{5}\ \text{W}$ — 1

2

Explanation: The resistance per metre multiplied by the length gives the total resistance.

(b) If the resistivity of copper is $1.67 \times 10^{-8}\ \Omega\ \text{m}$, calculate the diameter of the cable. [3]

Answer

$\frac{R}{l} = \frac{\rho}{A} \Longrightarrow A = \frac{1.67 \times 10^{-8}}{1.50 \times 10^{-4}} = 1.11 \times 10^{-4}\ \text{m}^2$ — 1

$A = \frac{\pi d^2}{4} \Longrightarrow d = \sqrt{(\frac{4A}{\pi})}$ — 1

$d = 1.19 \times 10^{-2}\ \text{m}$ — 1

3

Explanation: The resistance per unit length equals resistivity divided by area.

(c) Describe the dependence of resistivity on temperature for metal. [2]

Answer

The resistivity of a metal increases (approximately linearly) with temperature — 1

because the lattice ions vibrate with larger amplitude, so free electrons collide more often and the mean free time decreases (the number of free electrons stays the same) — 1

2

Total for question **7**

Explanation: $\rho = \frac{m}{ne^2\tau}$: in a metal n is constant, so ρ rises as τ falls.

22. An ideal capacitor of capacitance 3300 μF and a resistor of resistance 1.0 Ω are connected in series to an a.c. supply of frequency 50 Hz.

(a) Determine the impedance of the circuit. [4]

Answer

$X_C = \frac{1}{2\pi fC}$ 1

$X_C = \frac{1}{2\pi(50)(3300 \times 10^{-6})} = 0.965\ \Omega$ 1

$Z = \sqrt{(R^2 + X_C^2)}$ 1

$Z = \sqrt{(1.0^2 + 0.965^2)} = 1.39\ \Omega$ 1

4

Explanation: Resistance and reactance are 90° out of phase, so they combine like the sides of a right-angled triangle.

(b) Determine the phase difference between the current in the circuit and the voltage of the a.c. supply. [2]

Answer

$\tan\varphi = \frac{X_C}{R} = \frac{0.965}{1.0} \Longrightarrow \varphi = 44.0°$ (0.768 rad) 1

The current leads the supply voltage 1

2

Explanation: In an *RC* circuit the current leads the voltage by an angle between 0 and 90°.

(c) State **one** advantage of using an ideal capacitor. [1]

Answer

An ideal capacitor dissipates no power (no energy is lost as heat), so it limits the current without wasting energy 1

1

Total for question **7**

Explanation: Energy stored in the capacitor in one quarter-cycle is returned to the supply in the next.

Section C [26 marks]

23. (a) State **two** differences between electromotive force and electric potential difference. [2]

Answer

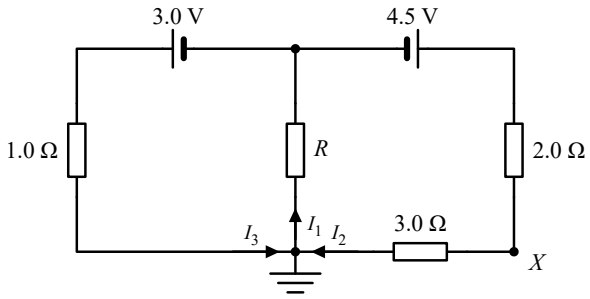
e.m.f. is the work done per unit charge by the source to move charge round the whole circuit (non-electrical → electrical energy); p.d. is the work done per unit charge between two points (electrical → other forms) — 1

An e.m.f. exists even when no current flows; a p.d. across a resistor exists only when current flows — 1

2

Explanation: Both are measured in volts (J C^{-1}), but they describe opposite energy conversions.

(b) Two batteries of e.m.f. 3.0 V and 4.5 V are connected as shown in the circuit diagram below.



The current I_1 is two times of current I_2.

(i) Determine the resistance R and current I_2. [5]

Answer

Junction rule: $I_1 = I_2 + I_3$ and $I_1 = 2I_2 \Longrightarrow I_3 = I_2$ — 1

Taking the earthed junction as 0 V, the potential of the top junction: via the left branch $V = 1.0I_3 - 3.0$; via the right branch $V = 5.0I_2 - 4.5$ — 1

$I_2 - 3.0 = 5.0I_2 - 4.5 \Longrightarrow I_2 = 0.375$ A — 1

$V = 0.375 - 3.0 = -2.625$ V $= -I_1R$, with $I_1 = 0.75$ A — 1

$R = \dfrac{2.625}{0.75} = 3.5\ \Omega$ — 1

5

Explanation: Using potentials from the earth point is equivalent to applying Kirchhoff's loop rule to each loop. The right branch contains 2.0 Ω + 3.0 Ω in series.

(ii) Determine the electric potential at X. [3]

Answer

I_2 flows from X to earth through the 3.0 Ω resistor	1
$V_X - 0 = I_2(3.0) = (0.375)(3.0)$	1
$V_X = +1.13$ V	1
	3

Explanation: Current flows from higher to lower potential through a resistor, so X is above earth potential.

(c) Explain the working principle of a shunt. [3]

Answer

A shunt is a low resistance connected in parallel with a galvanometer	1
Most of the current passes through the shunt; only the full-scale current I_g passes through the galvanometer, so a large current can be measured (an ammeter)	1
$I_g R_g = (I - I_g)R_s$; the small total resistance hardly changes the circuit current	1
	3
Total for question	**13**

Explanation: The shunt and galvanometer have the same p.d. because they are in parallel.

24. (a) State **two** differences between electric force and magnetic force that act on a charged particle. [2]

Answer

The electric force acts on a charge whether it is moving or at rest; the magnetic force acts only on a moving charge (not moving parallel to **B**)	1
The electric force is along **E** and can change the speed (does work); the magnetic force is perpendicular to **v** and **B**, changing only the direction (does no work)	1
	2

Explanation: $\mathbf{F} = q\mathbf{E}$ and $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$.

(b) In Thomson's experiment, an electron is accelerated by a high voltage into a region of an electric field and magnetic field of 92 V m^{-1} and 0.020 T respectively. The electron is then

deflected in the absence of electric field. Determine the velocity and the radius of deflection of the electron. [7]

Answer

Undeflected (velocity selector): $eE = evB$ — 1

$v = \frac{E}{B} = \frac{92}{0.020}$ — 1

$v = 4.6 \times 10^3 \text{ m s}^{-1}$ — 1

In **B** only: $evB = \frac{mv^2}{r}$ — 1

$r = \frac{mv}{eB}$ — 1

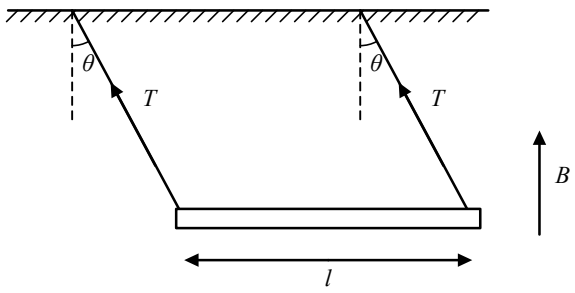
$r = \frac{(9.11 \times 10^{-31})(4.6 \times 10^3)}{(1.60 \times 10^{-19})(0.020)}$ — 1

$r = 1.31 \times 10^{-6} \text{ m}$ — 1

7

Explanation: With both fields present the electric and magnetic forces balance; when the electric field is removed, the magnetic force alone provides the centripetal force.

(c) A wire of length l and weight mg is hanged on two strings of tension T. The wire carries a current I and deflected from the vertical at an angle θ in the presence of uniform magnetic field, B, as shown in the diagram below.



Derive an expression for the magnetic field strength. [4]

Answer

Forces on the wire: weight mg down, tension $2T$ along the strings, magnetic force BIl horizontal — 1

Vertical: $2T \cos \theta = mg$ — 1

Horizontal: $2T \sin \theta = BIl$ — 1

$\tan \theta = \frac{BIl}{mg} \Longrightarrow B = \frac{mg \tan \theta}{Il}$ — 1

4

Total for question **13**

Explanation: The field is vertical and the current horizontal, so the magnetic force is horizontal and pushes the wire sideways until the three forces balance.